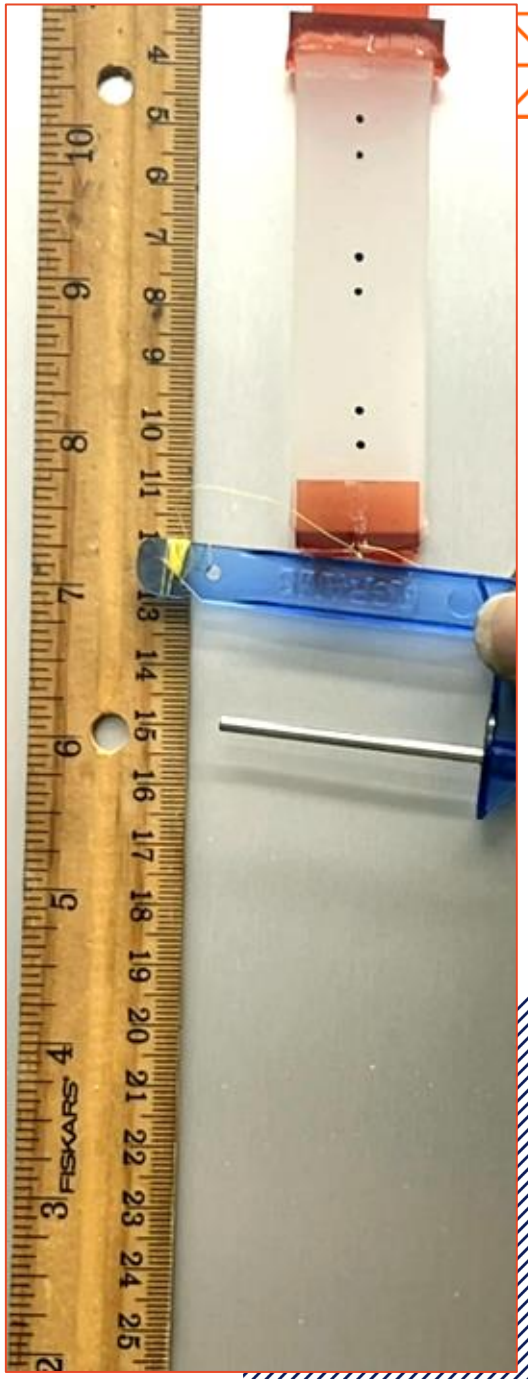
# Analysis of “super stretchy” behavior of Ecoflex silicone rubber

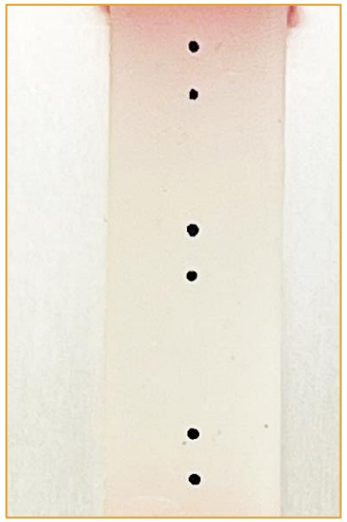
## Final project

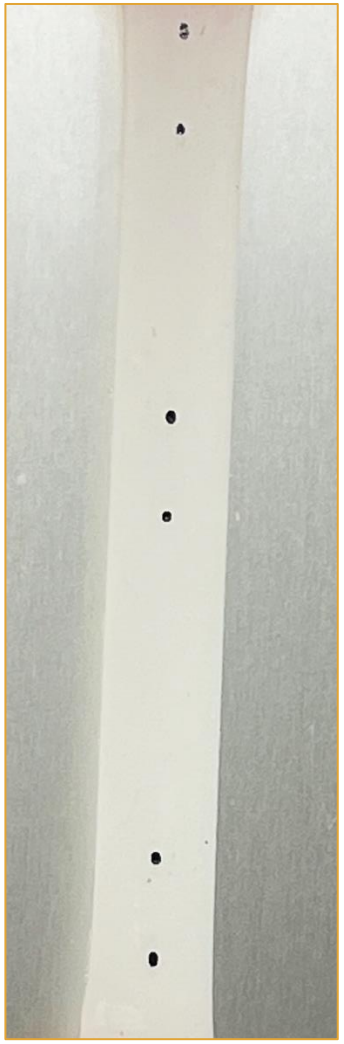
*Presented by,*
*Srividhya Sridhar*



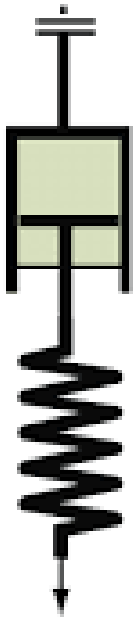
MechSE
TAM 534 - Spring 2025

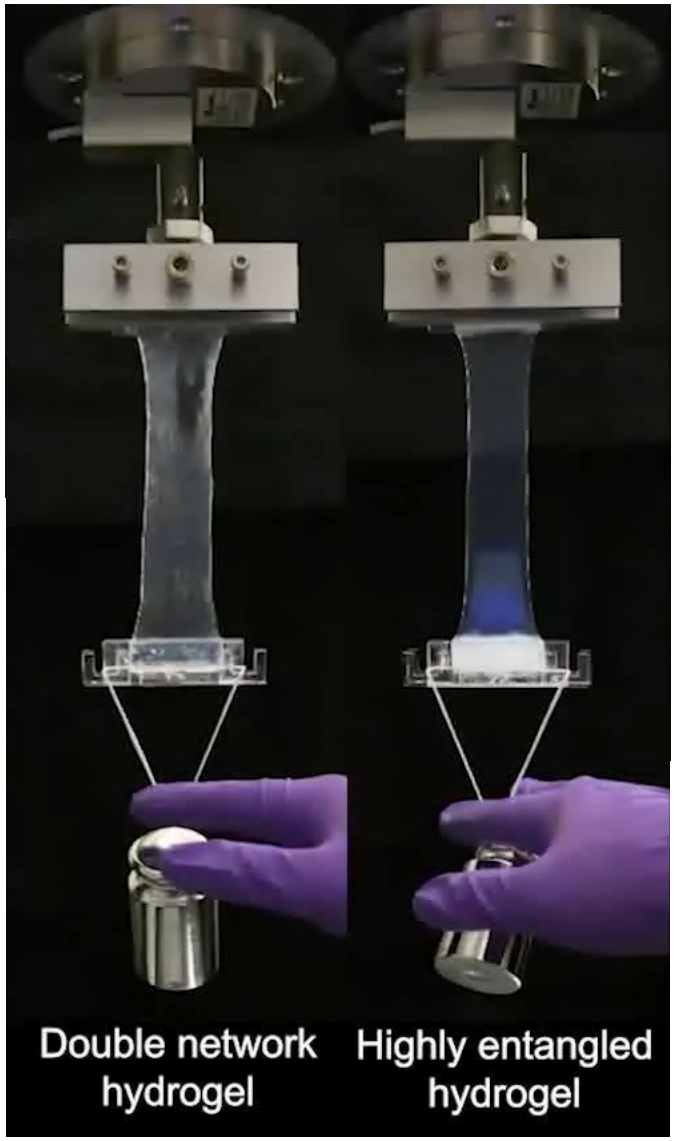
# Ecoflex is a silicone rubber – soft, stretchy and very tough.



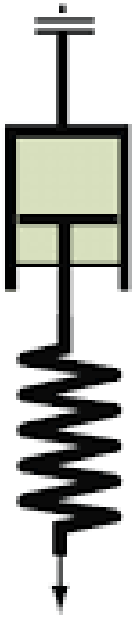


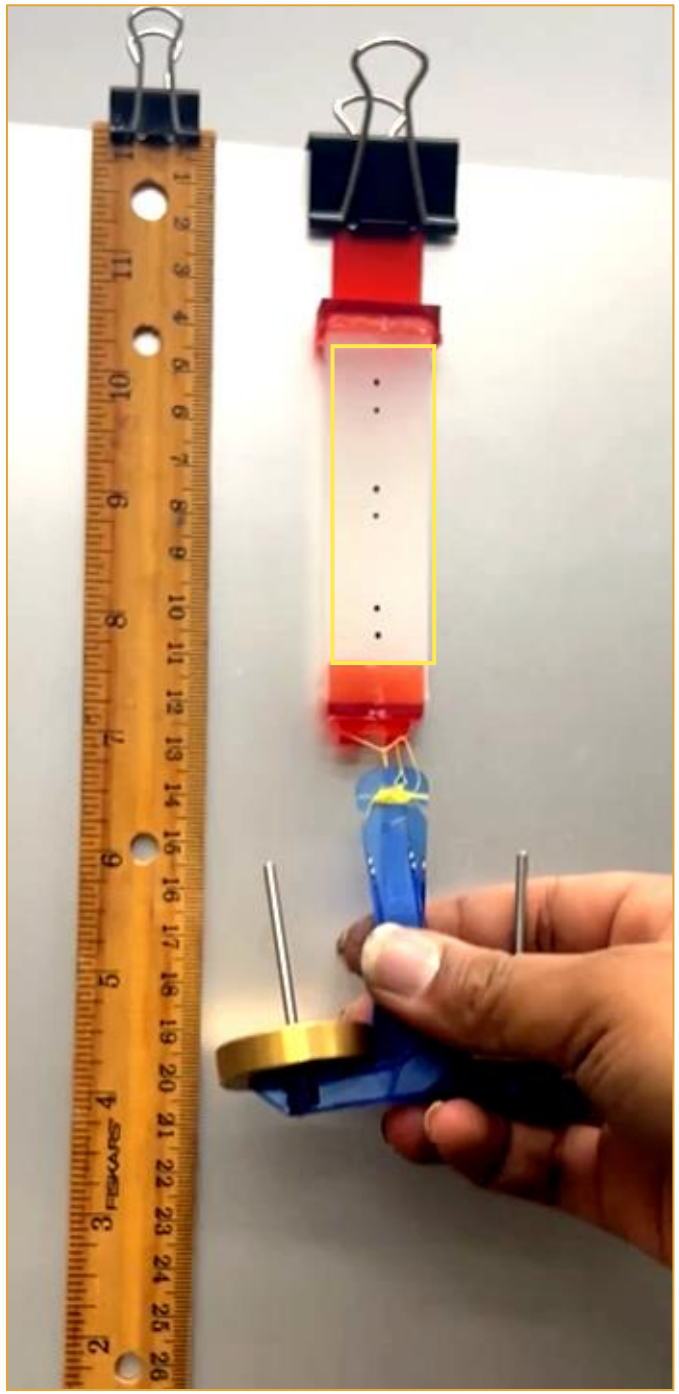
Kim et al., *SCIENCE (2021)*



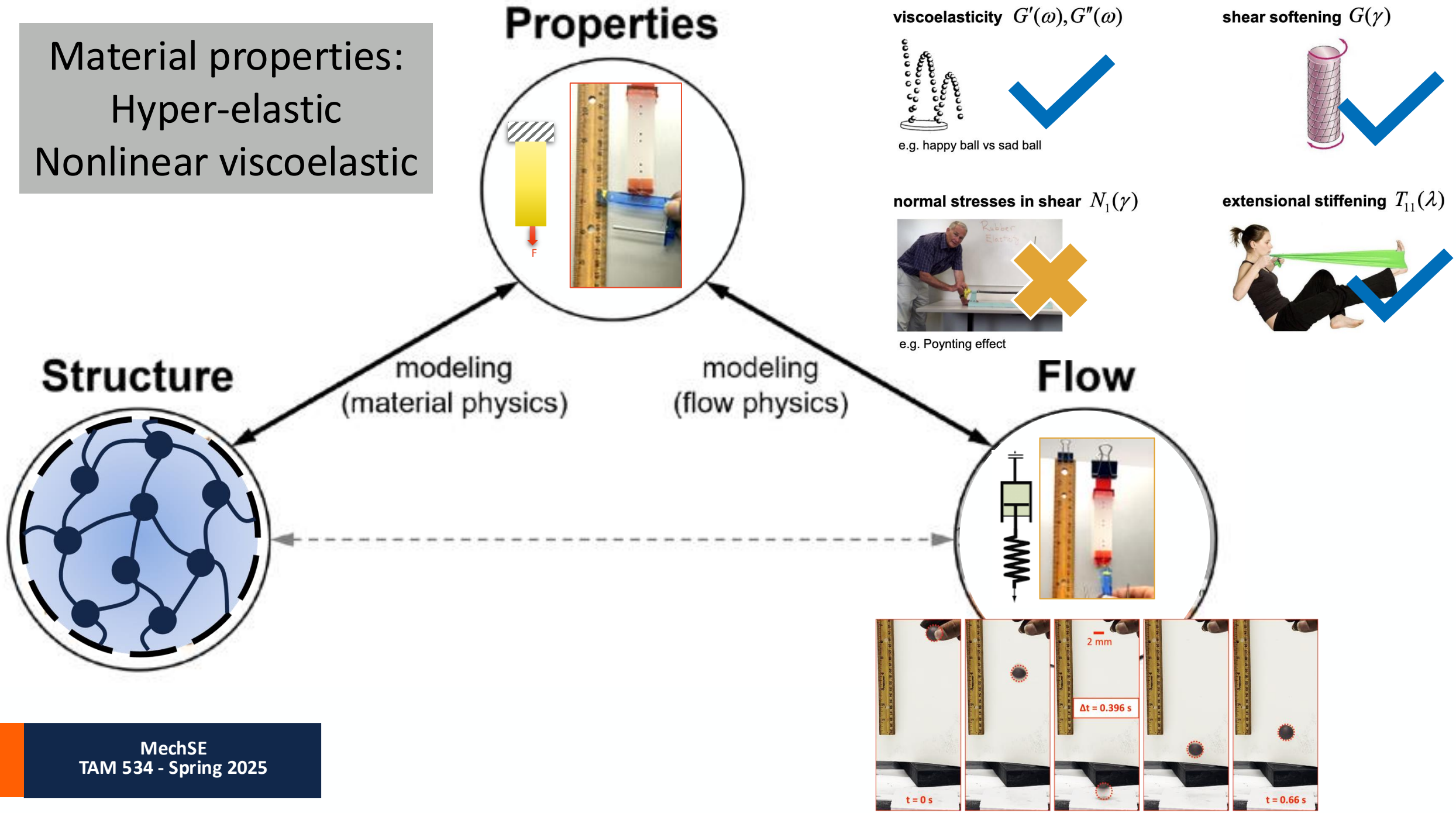


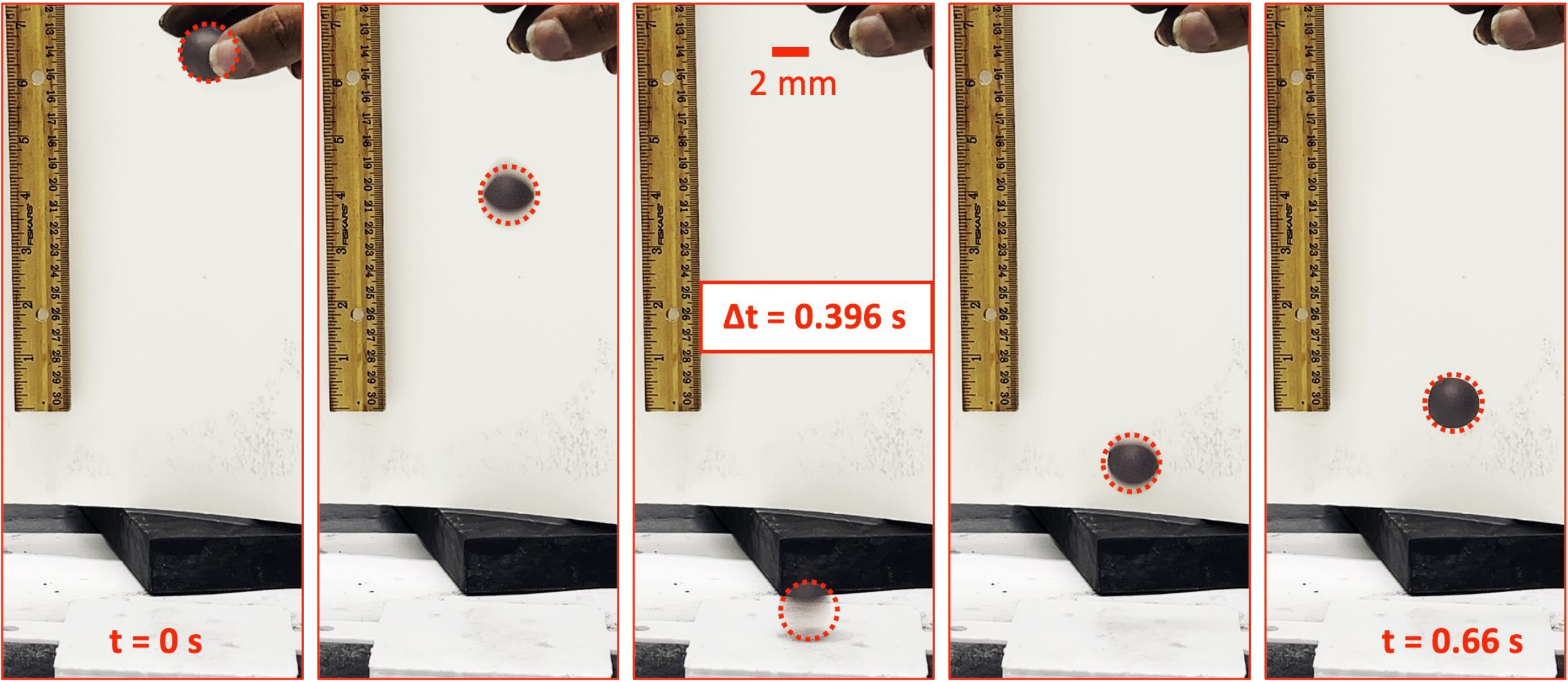
We have a viscoelastic material





# Material properties: Hyper-elastic Nonlinear viscoelastic

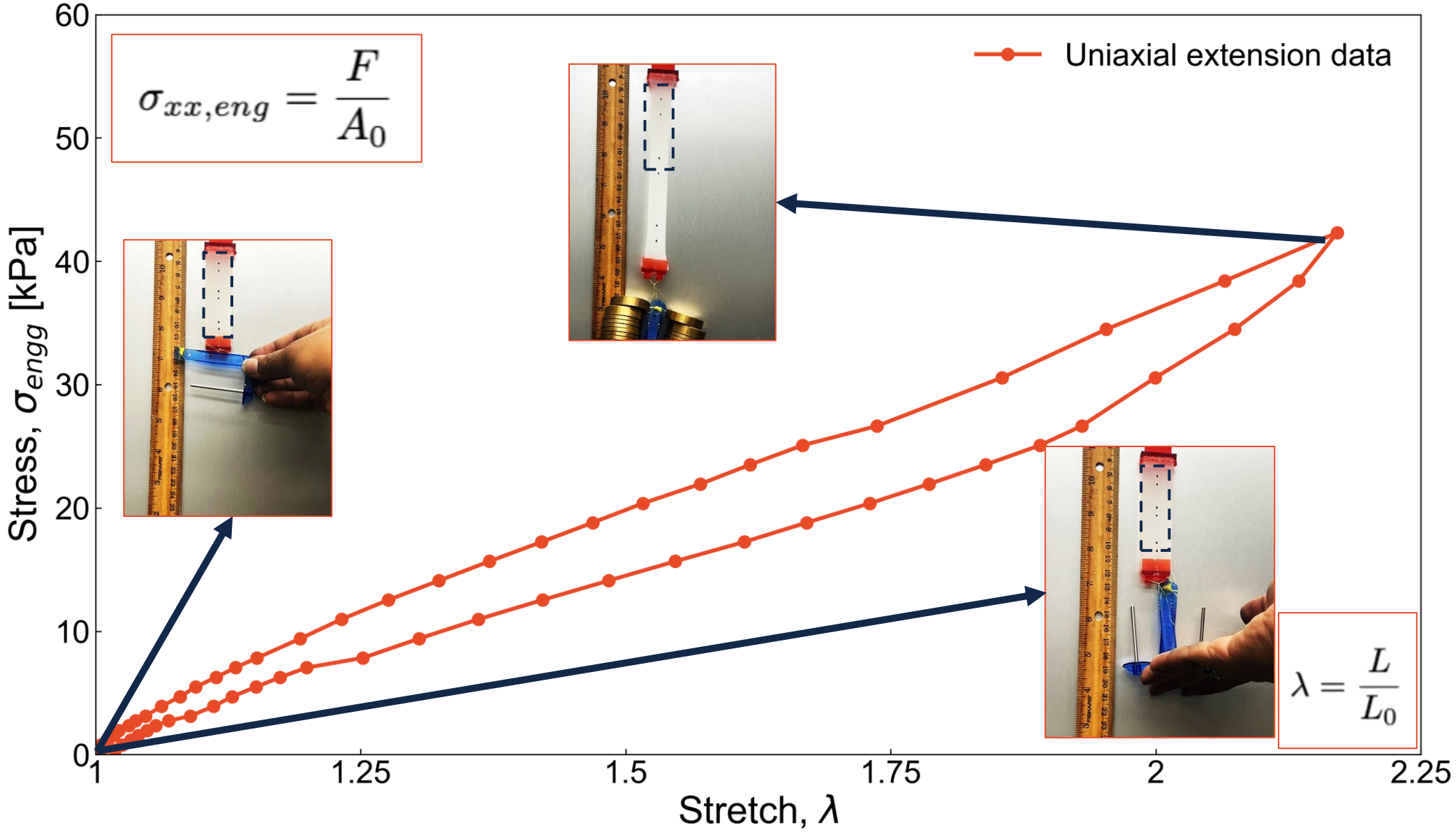


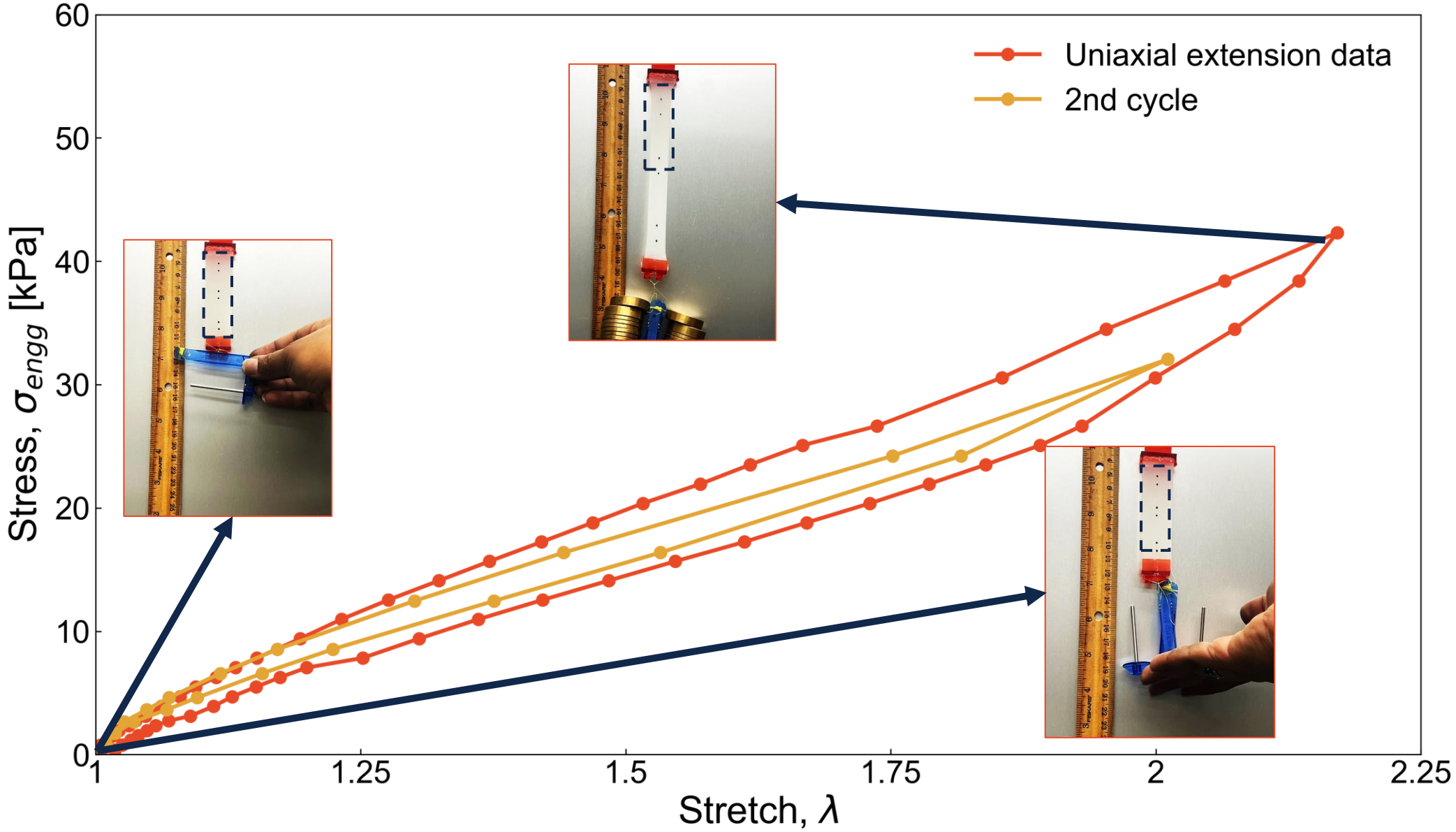



$\tan\delta \sim 0.34$

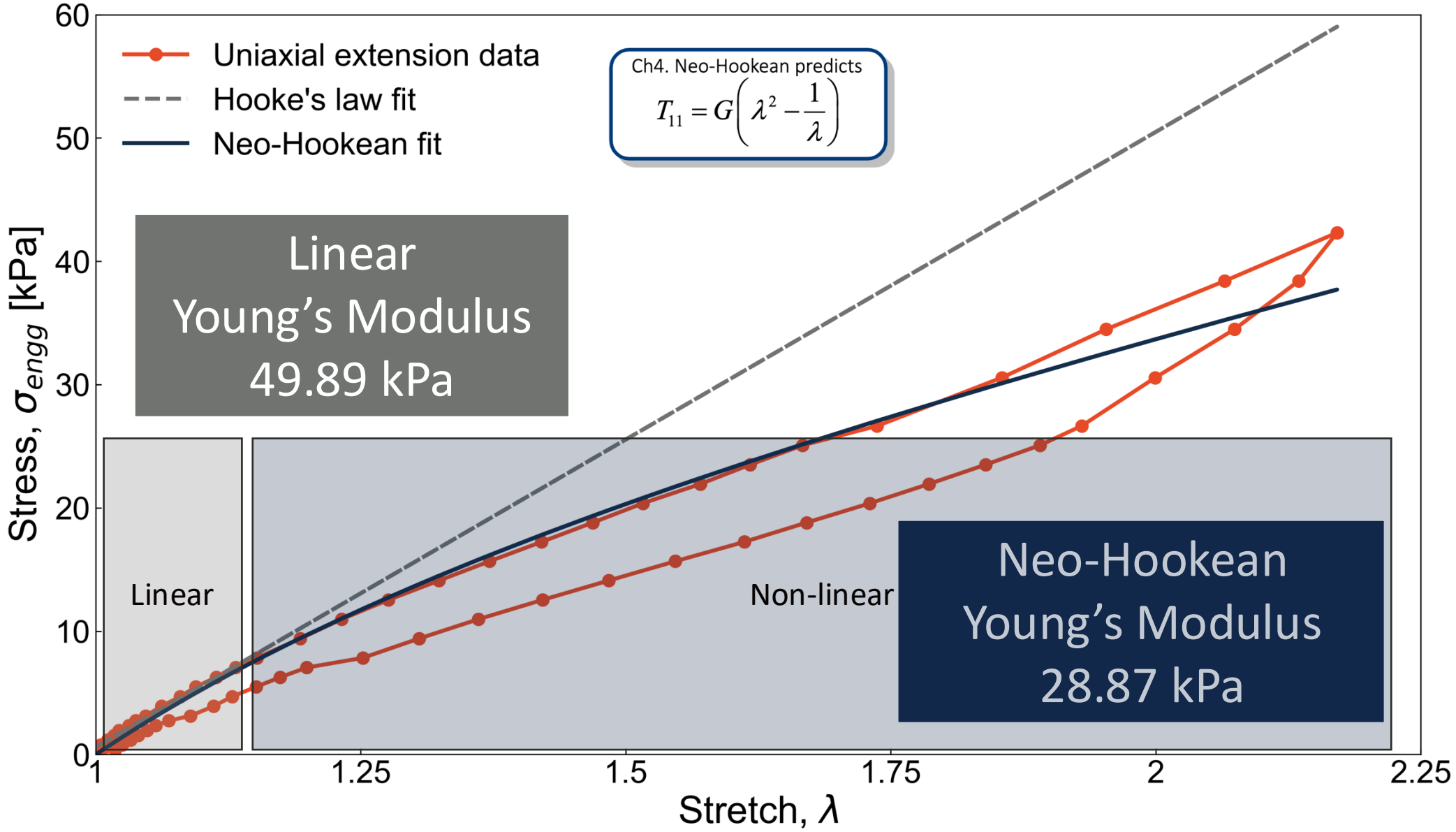
$h_{reb} < h_{drop}$

Ecoflex
Applied load


$\sigma_{xx,eng} = \frac{F}{A_0}$
Uniaxial extension data
Stress, $\sigma_{engg}$ [kPa]
Stretch, $\lambda$
0
10
20
30
40
50
60
1
1.25
1.5
1.75
2
2.25
$\lambda = \frac{L}{L_0}$



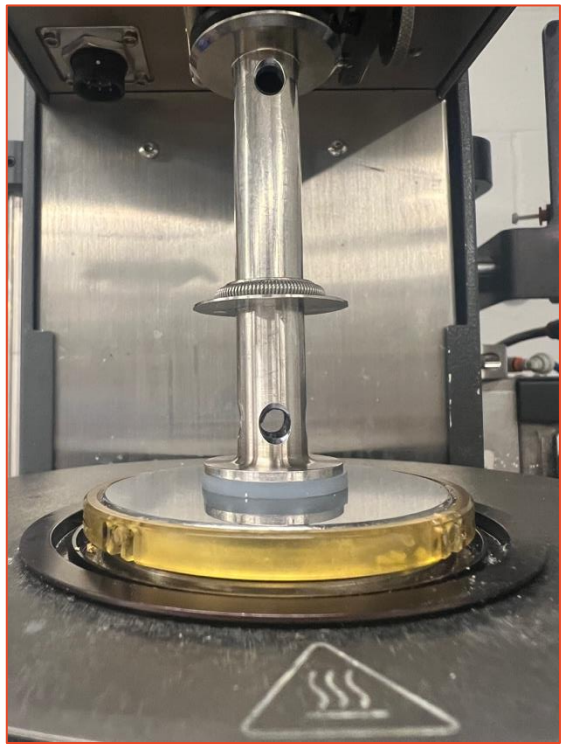

**Ecoflex exhibits Mullin's damage – which is a shear softening behavior!**

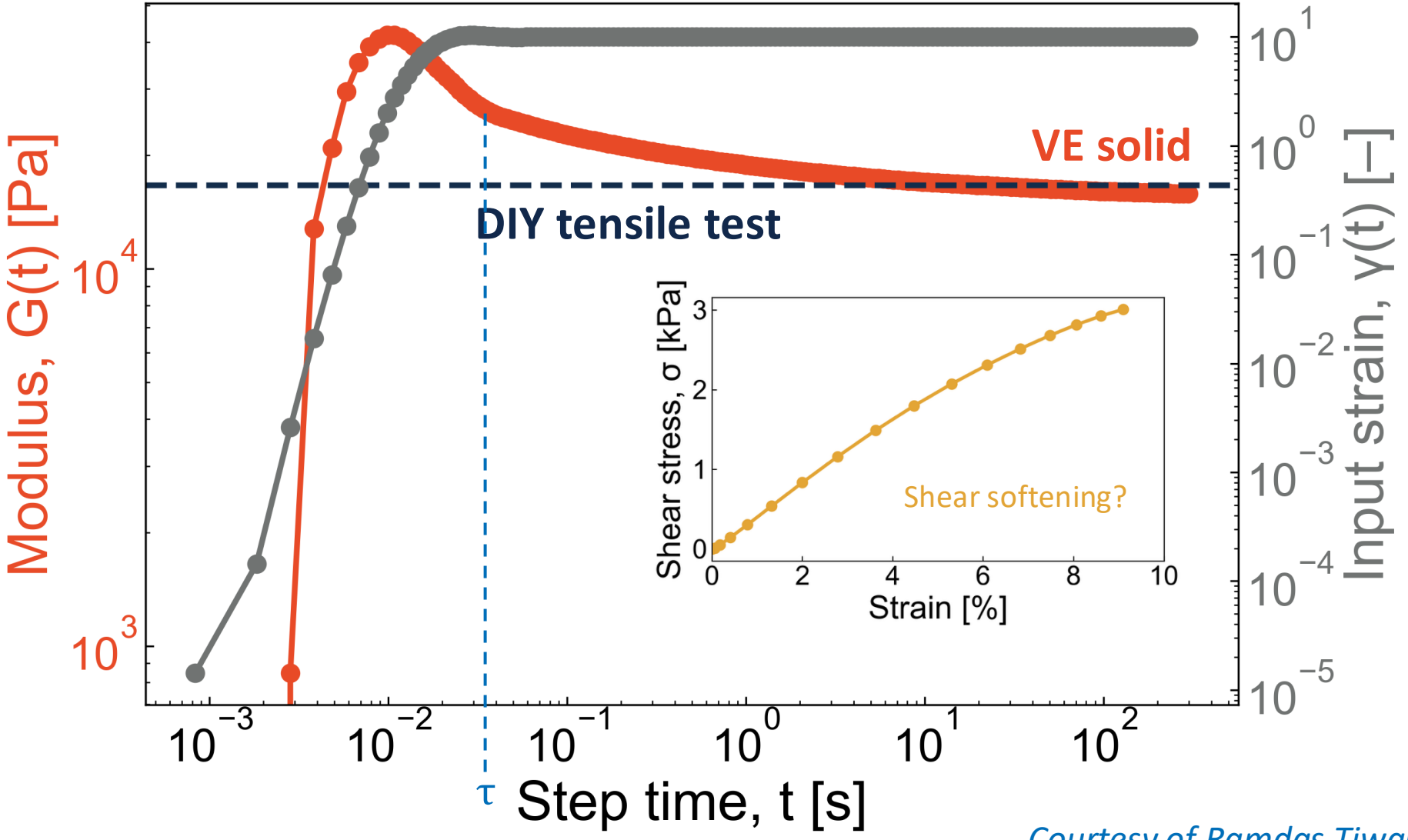



# Elastic modulus value agree with literature!

*J. Mater. Chem. C*, 2020,**8**, 1273-1279

# Similar elastic modulus obtained from stress relaxation experiment using Rheometer!

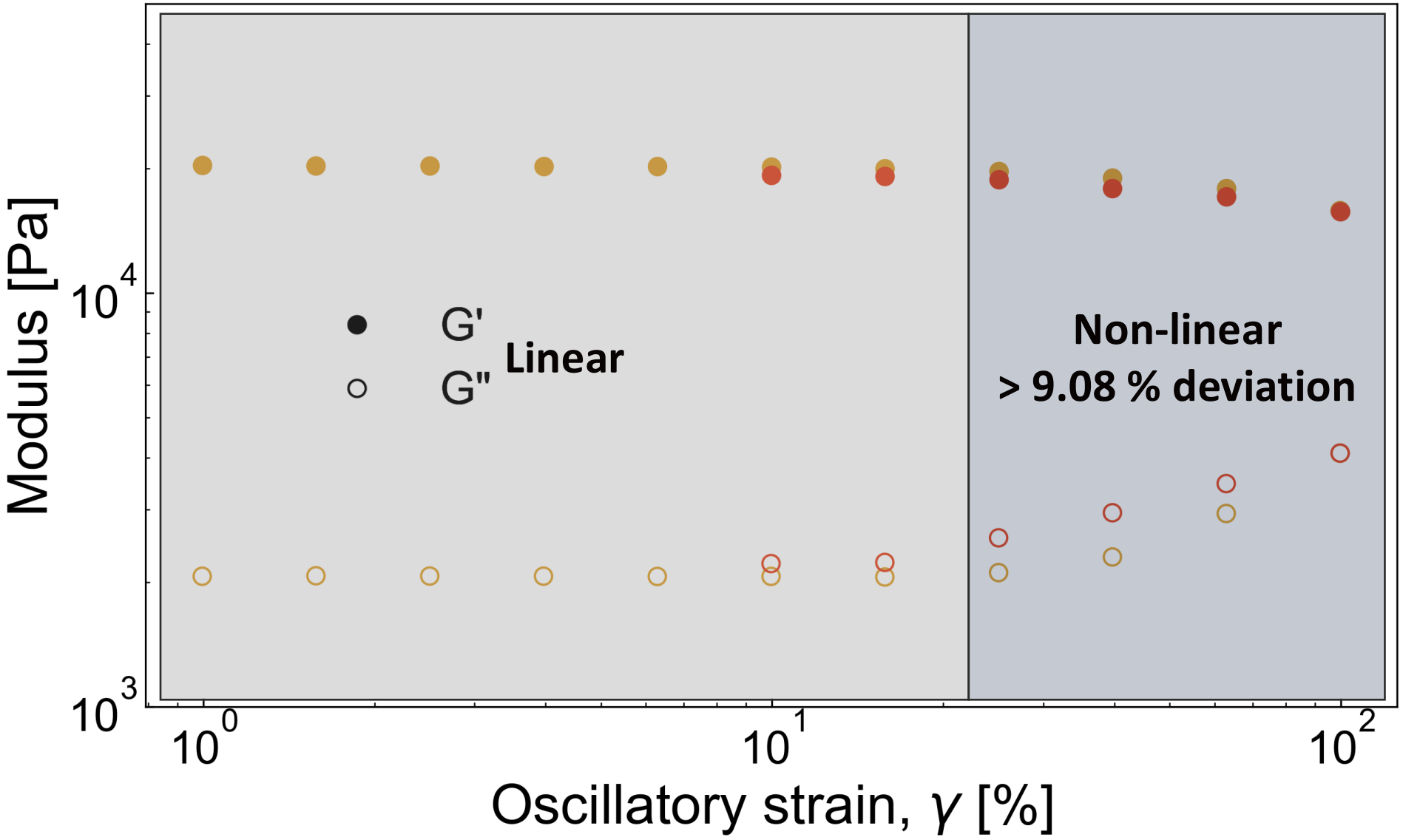





*Courtesy of Ramdas Tiwari*

# Amplitude sweep test on Rheometer reveal the onset of nonlinearity!

Loss is increasing with the 2nd test – perhaps Mullin's damage is seen here too!



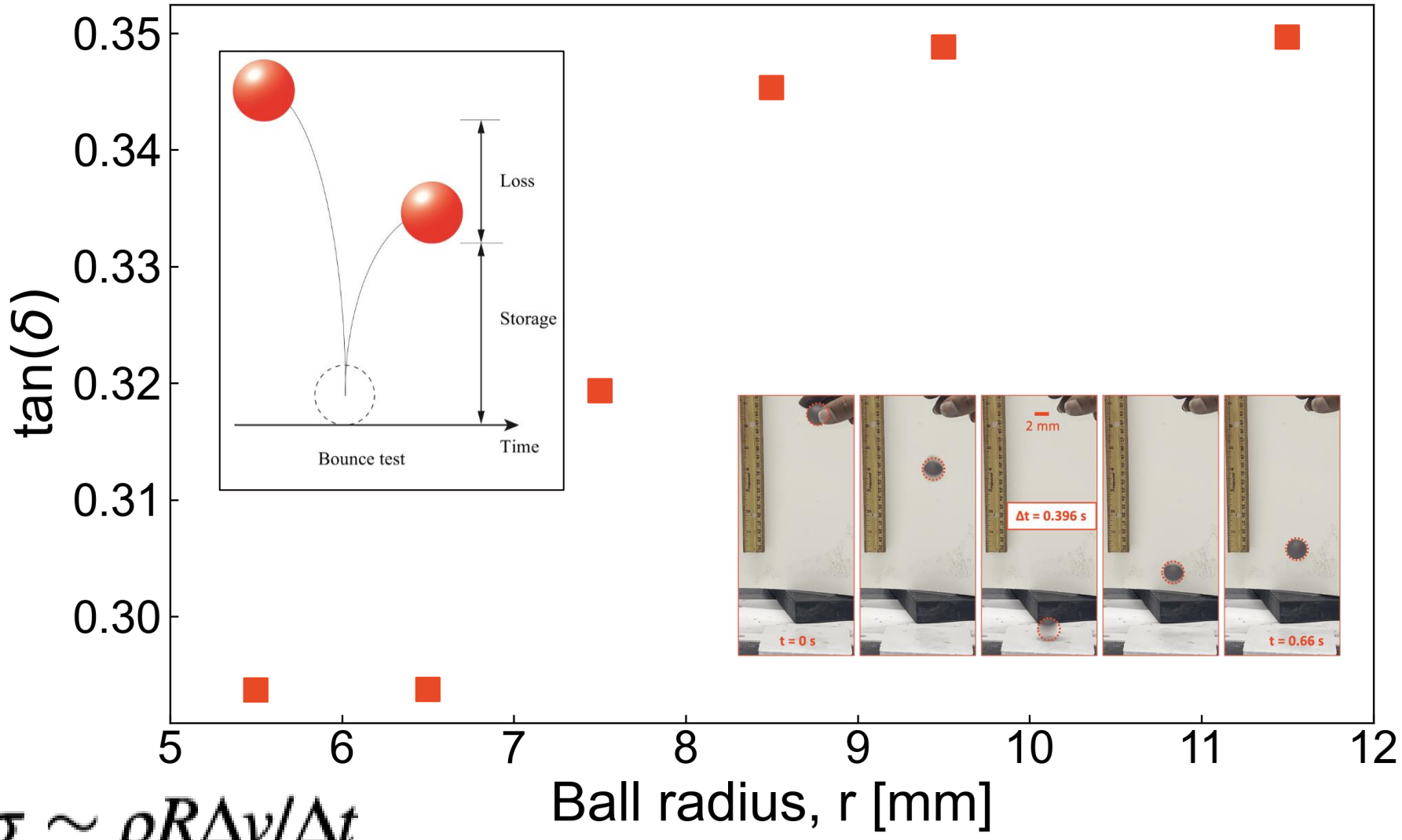



*Courtesy of Ramdas Tiwari*

# Tan delta appears to be increasing with increase in forcing strengthx

$$\tan\delta = \frac{1}{\pi}\ln\left(\frac{1}{f}\right)$$

$$f = h_{reb}/h_{drop}$$

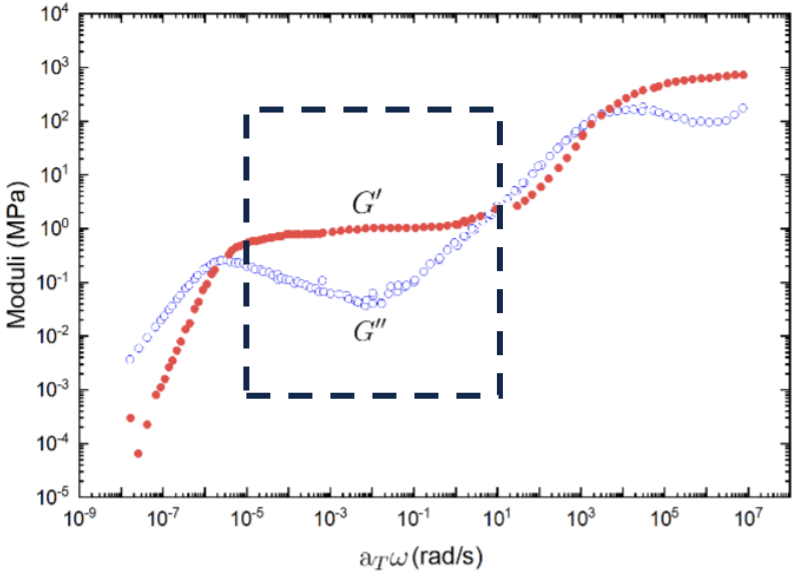


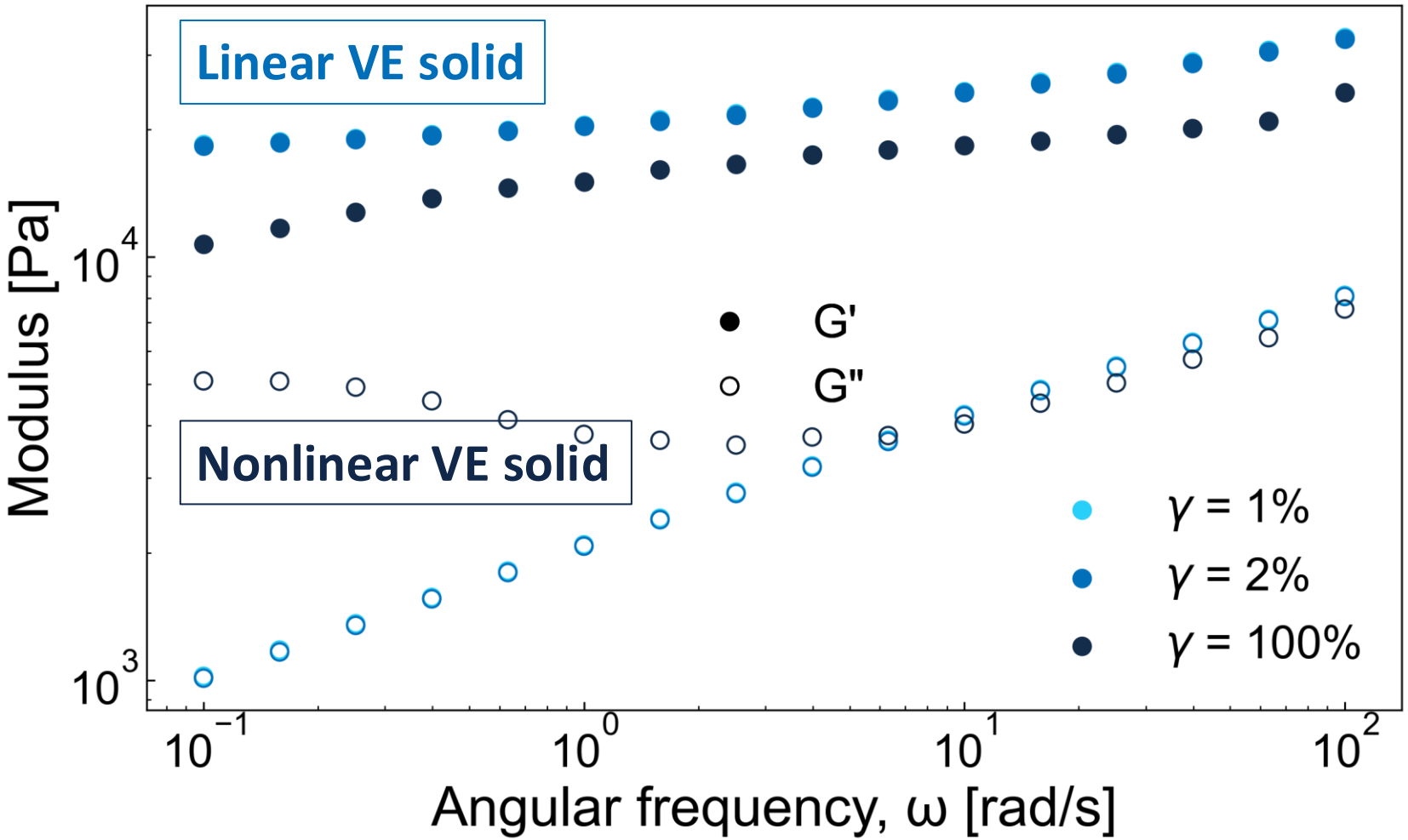

$$\sigma \sim \rho R \Delta v / \Delta t$$



*J. Rheol. 68, 113–144 (2024)*

# SAOS data shows viscoelastic solid "Rubbery" behavior.

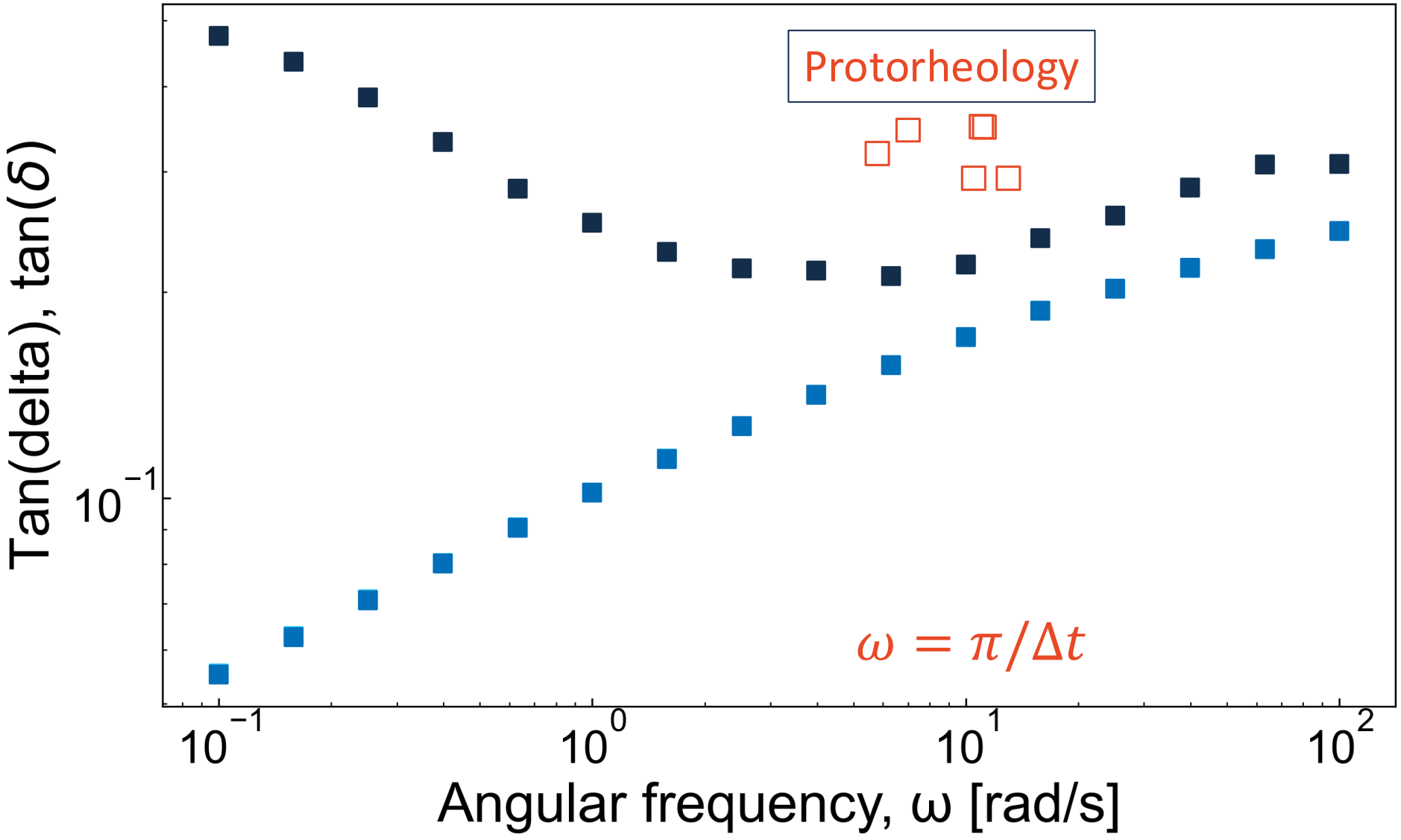





*Courtesy of Ramdas Tiwari*

**Impact time was calculated for each bounce tests of varying sizes and calculated the frequency.**

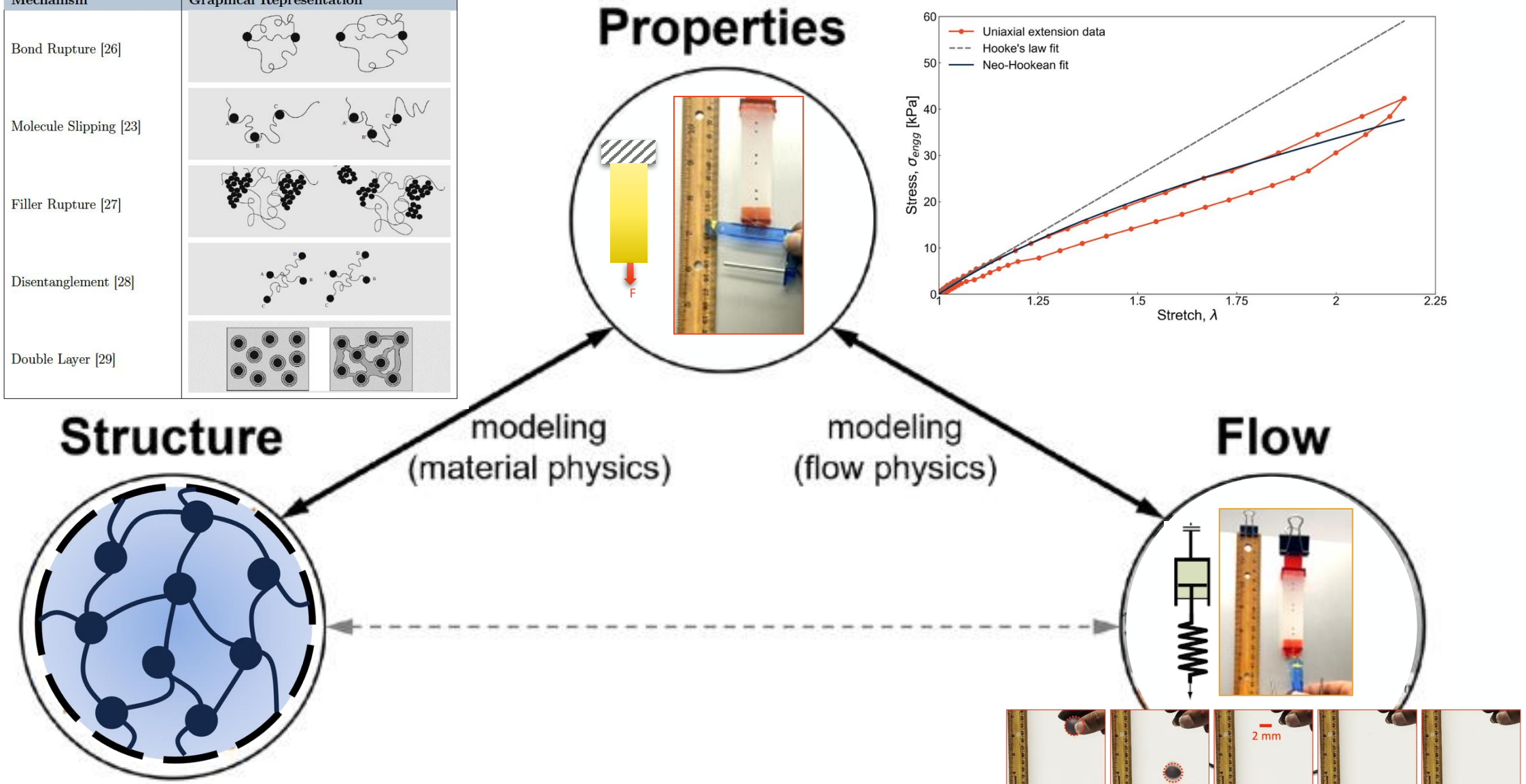



*Perhaps my forcing strength was nonlinear*

*J. Rheol. 68, 113–144 (2024)*

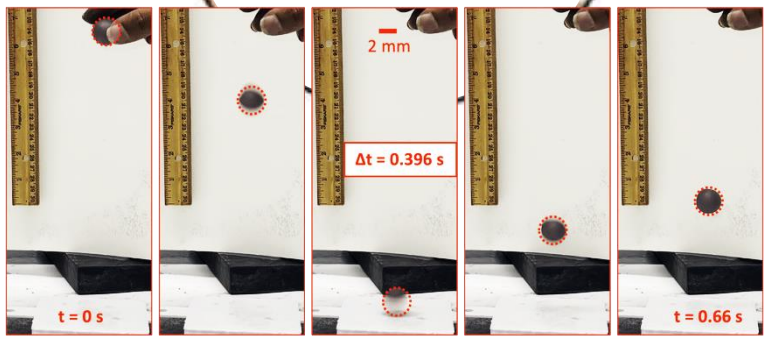
# Mullin's damage mechanisms

| Mechanism | Graphical Representation |
|---|---|
| Bond Rupture [26] |  |
| Molecule Slipping [23] |  |
| Filler Rupture [27] |  |
| Disentanglement [28] |  |
| Double Layer [29] |  |

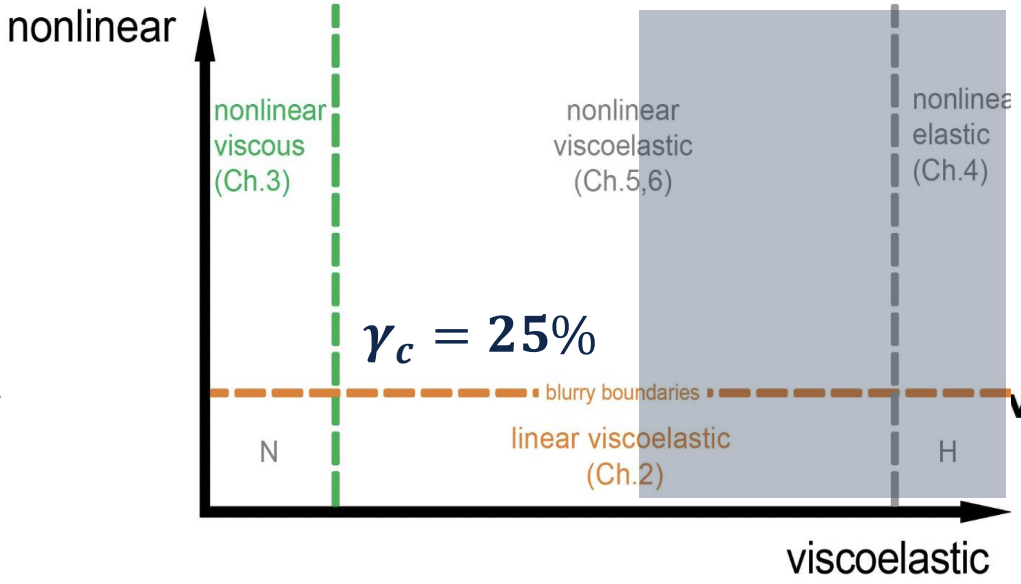




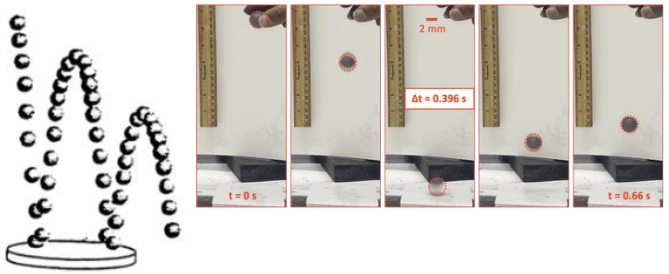




# 3 out of 4 key phenomena in the solid polymer are demonstrated

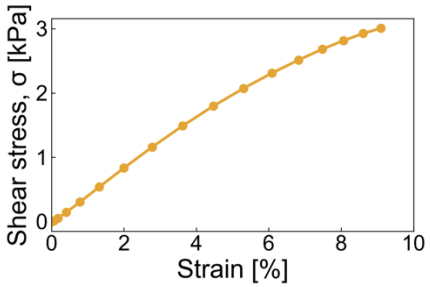



**viscoelasticity** $G'(\omega), G''(\omega)$

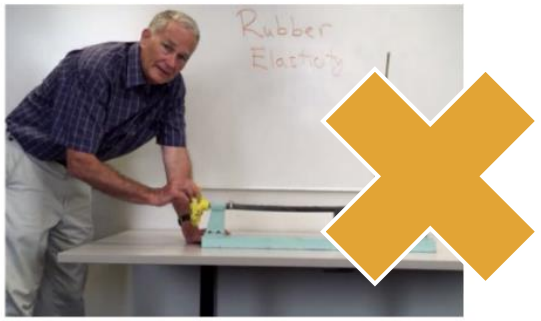


e.g. happy ball vs sad ball

**shear softening** $G(\gamma)$



**normal stresses in shear** $N_1(\gamma)$

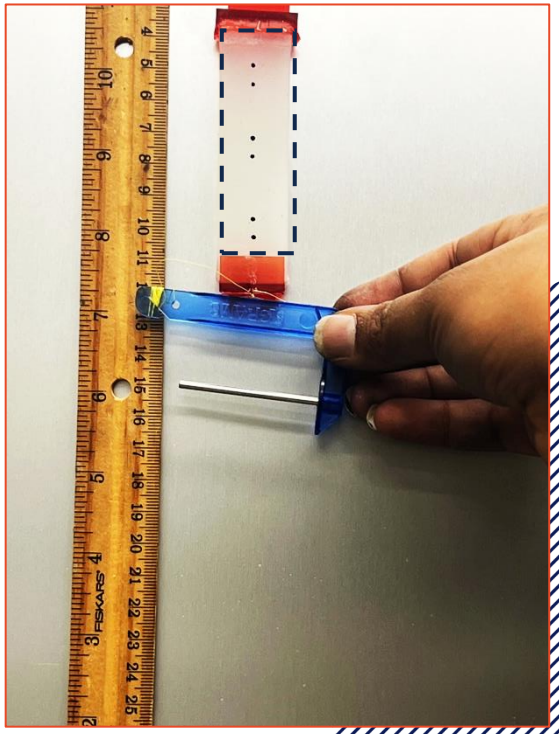



e.g. Poynting effect

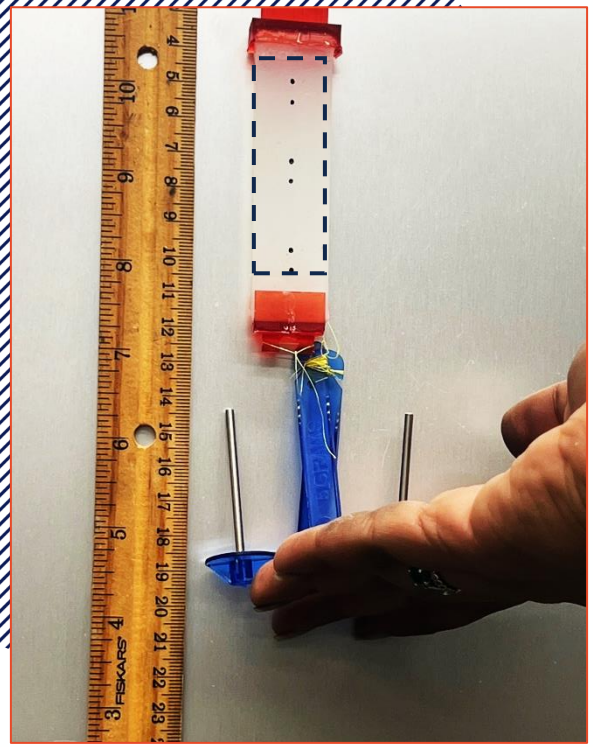
**extensional stiffening** $T_{11}(\lambda)$

Nonlinear behavior under uniaxial extension

# Takeaways



Irreversible dissipation



After loading - unloading

***Ecoflex exhibits Mullin's damage and highly nonlinear behavior***

- DIY tests – bouncing ball tests and Extensional tests were done.
- 3 out of 4 key phenomena was revealed from the videos and the images.
- Material behavior was quantitively described from the DIY experiments as well as the Rheometer experiments.

# THANK YOU!

ILLINOIS

Srividhya
Sridhar